

effect. Introducing the use of a ROPE into this decision making process shows some promise, at least for avoiding Lindley cases and some other pitfalls of point-null testing.

These suggestions are not the only ways that the problems raised here can be resolved, and it is beyond the scope of this paper to mount an exhaustive survey of potential resolutions. Instead, the main point of this paper has been the point out that there is an additional Achilles' heel in the Bayesian framework besides the choice of priors. Bayesian hypothesis testing also is vulnerable to attempting to "prove" a point-valued null, and that vulnerability becomes more apparent with larger samples.

References

- Bolstad, W. M., & Curran, J. M. (2016). *Introduction to Bayesian statistics, 3rd Ed.* John Wiley & Sons.
- Dienes, Z. (2014). Using Bayes to get the most out of non-significant results. *Frontiers in psychology*, 5, 781.
- Gallistel, C.R. (2009) The importance of proving the null. *Psychological review*; 116(2): 439-453.
- Gelman, A., Carlin, J. B., Stern, H. S., & Rubin, D. B. (2004). *Bayesian data analysis 2nd Ed.* London: Chapman and Hall.
- Jeffreys, H. (1961). *Theory of probability 3rd Ed.* New York: Oxford University Press.
- Konijn, E. A., van de Schoot, R., Winter, S. D., & Ferguson, C. J. (2015). Possible solution to publication bias through Bayesian statistics, including proper null hypothesis testing. *Communication methods and measures*, 9(4), 280-302.
- Kruschke, J. K. (2011). Bayesian assessment of null values via parameter estimation and model comparison. *Perspectives on psychological science*, 6, 299–312.
- Kruschke, J. K. (2018). Rejecting or accepting parameter values in Bayesian estimation. *Advances in methods and practices in psychological science*, 1 (2), 270–280.

- Lindley, D. V. (1957). A statistical paradox. *Biometrika*, 44, 187–192.
- Liu, C. C., & Aitkin, M. (2008). Bayes factors: Prior sensitivity and model generalizability. *Journal of mathematical psychology*, 56, 362–375.
- Lynch, S. M. (2007). *Introduction to applied Bayesian statistics and estimation for social scientists*. Springer Science & Business Media.
- Masson, M. E. (2011). A tutorial on a practical Bayesian alternative to null-hypothesis significance testing. *Behavior research methods*, 43(3), 679-690.
- Morey, R. D., Romeijn, J. W., & Rouder, J. N. (2016). The philosophy of Bayes factors and the quantification of statistical evidence. *Journal of mathematical psychology*, 72, 6-18.
- Robert, C. P. (2014). On the Jeffreys-Lindley paradox. *Philosophy of science*, 81(2), 216-232.
- Rouder, J. N., Haaf, J. M., & Vandekerckhove, J. (2018). Bayesian inference for psychology, part IV: Parameter estimation and Bayes factors. *Psychonomic bulletin & review*, 25(1), 102-113.
- Rouder, J. N., Speckman, P. L., Sun, D., Morey, R. D., & Iverson, G. (2009). Bayesian t tests for accepting and rejecting the null hypothesis. *Psychonomic bulletin & review*, 16(2), 225-237.
- Wagenmakers, E.-J. (2007). A practical solution to the pervasive problem of p values. *Psychonomic bulletin and review*, 14, 779–804.
- Wagenmakers, E.-J., Gronau, Q.F., Dablander, F., & Etz, A. (2019). *The support interval*. ArXiv e-prints.